

Edgecumbe Barge 26' ABS Inspection



COVER SHEET – STATEMENT OF WORK (SOW)

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Statement of Work (SOW)

1.0 GENERAL REQUIREMENTS: This document invokes the requirements applicable to the docking and maintenance work to be performed on Edgumbe RCD Barge. Work performed and accomplished shall be in accordance with the highest quality workmanship standards and practices commonly accepted in the commercial marine industry and the technical references listed herein. The Contractor is responsible for understanding and complying with all requirements specified herein. The requirements listed herein work in conjunction with the technical drawings and other references to describe the essential features, functions, and requirements of the work performed.

1.1 Vessel Characteristics:

Edgumbe RCD Barge		Built by: Zidell Dismantling Inc.	
Length:	240 FT	Draft:	2' 11" FT
Width:	60 FT	Full Displacement:	928 L Tons
Age:	48 YRS		
Hull Type:	Steel		

1.2 Scope Overview: Coordinate with the Contracting Officer, via the Contracting Officer's Representative (COR) to deliver the vessel to the Contractor's facility. Upon docking vessel, wash, and clean vessel. Accomplish zinc anode replacement; preliminary and final air tests of all tanks/voids; tank/void open, clean and gas free; and support ABS and USCG inspections. Undock the vessel. Arrange and coordinate the delivery of the vessel back to the Government with the Contracting Officer via the COR.

1.3 Definitions: As used throughout this Specification, the following definitions and terms have meanings as indicated:

1.3.1 Unless otherwise stated, "as shown", "as indicated", "as detailed", or other words of similar import refer to the contractual documents including reference documents and publications.

1.3.2 The words “as directed”, “as approved”, “when authorized” or words of similar import refer to the direction, requirements, permission, approval, or acceptance by the Contracting Officer.

1.3.3 The term “Objective Quality Evidence” (OQE) shall mean current, accurate, and complete objective documentation and information submitted together as a complete data package sufficient that a qualified and disinterested third party could analyze and evaluate whether technical requirements were achieved.

1.3.4 Where “in accordance with Regulatory Body requirements” is stated, the industrial efforts shall be in accordance with the appropriate commercial regulatory body requirements unless otherwise authorized by the Government.

1.3.5 The term “marine” or “marine quality” means an item constructed of materials unaffected by moisture, sea spray, extremes in temperature or other hazards of the marine environment and designed and constructed to perform its intended functions under marine operating conditions.

1.3.6 The term “herein” means the applicable section(s) of this Specification.

1.3.7 The term “NAVSEA” means the Naval Sea Systems Command, one of five system commands, responsible for engineering, building, supporting, and maintaining the Navy’s fleet of vessels.

1.3.8 The term “Stakeholder” means customer, Government, and any agency conducting work during the contracted availability timeline.

1.3.9 The term “SUPERVISOR” is defined as the local Government activity responsible for the execution and contract administration of Navy maintenance and modernization work. For the purposes of this SOW this activity is the assigned Code 101 representative.

1.3.10 The term “OEM” or Original Equipment Manufacturer is a company whose goods are used as components in the products of another company. An OEM creates parts and components that are used by other companies in their finished products.

1.3.11 The term “AIT” or Alteration Installation Team is defined as a Navy team (military, civilian or civilian contractor) under the direct supervision of

Navy personnel, that is trained and equipped to accomplish specific Alterations or temporary modifications on specified ships.

1.3.12 Worksite is defined as within the specific compartment or space where physical productive work is occurring. For work performed on Navy boats and craft or service craft, the “worksite” is defined as on or near the vessel.

1.3.13 Foreign Material Exclusion (FME) is used to maintain system cleanliness. Accomplishment cleanliness restoration, will be invoked within the work item when cleanliness is lost or suspected of being lost.

1.3.14 Key Event: An event that, if slippage occurs, could impact or delay the overall schedule, or prevent timely delivery of the vessel/asset. Key Events are identified by the contract, the SUPERVISOR, or the contractor.

1.3.15 Milestone: A significant event identified by the Maintenance Team. Milestones are used as a scheduling aid and establish significant points where progress must be evaluated and confirmed. Accumulated failure to achieve Milestones on schedule may result in missed Key Event. Milestones may be identified by either the contractor or the SUPERVISOR.

1.3.16 Critical Path: That sequence of Work Activities which forms the work and test chain of the longest duration and directly affects the completion of the availability. Factors that influence when a Work Activity is on the Critical Path include: time duration required for the Work Activity, space limitations, manpower available, and the predecessor/successor relationships between Work Activities. Typically, the Critical Path is determined by automated schedule analysis and will include any sequential set of Work Activities forming the longest chain of events extending throughout the schedule and which has the least Total Float.

1.3.17 Total Float: The total number of days that a path of Work Activities can be delayed without affecting the project finish date. A path of Work Activities is established by predecessor and successor relationships.

1.3.18 The acronym “GDO” stands for Government Docking Officer.

1.3.18.1 The GDO may be present at the time of docking a vessel/ asset using a floating or graving dock, or the vertical lift facility is raised and shall remain at the dock or vertical lift facility until all blocks are well covered to make sure that no blocks come adrift or are misplaced to endanger the ship when landing.

1.3.18.2 The GDO shall be notified and provided blocking plan, for any bump, shift of blocks, vessel/asset fleeting, or vessel/asset transition while within the Contractor's possession or facility. GDO will ensure calculations and block placement is consistent with engineering requirements, and blocks align with proper vessel/asset structure for the safety of the vessel/asset.

1.3.18.3 The GDO may be present at the time of undocking a vessel/ asset using a floating or graving dock, or the vertical lift facility is lowered ensure that no blocks or block caps remain affixed to the vessel/asset and the vessel/asset is seaworthy, without list, as not to endanger the ship.

1.3.19 The term "drydocking" refers to an evolution in which a vessel/asset is landed on blocks at a drydocking facility, in accordance with GDO-approved docking and blocking plans, enabling production work to be performed

1.3.20 The term "undocking" refers to an evolution in which a vessel/asset is removed from a drydocking facility for purposes returning a vessel to the water.

1.3.21 The term(s) "fleeting" including "bumping" or "shifting" refers to when a vessel/asset is unloaded and re-landed to the blocks without leaving a facility for purposes of docking or undocking other vessels or for shifting the position of the ship on the blocks for underbody coating system application or other production work.

1.4 **Arrival Conference:** The Government will arrange an arrival conference with the Contractor after contract award.

1.5 **Required Documents:** Administrative requirements of this SOW shall begin at contract award and be updated throughout the contract period.

1.5.1 Prepare and maintain a weekly updated schedule necessary for the purpose of establishing an orderly and systematic accomplishment of production work and material ordering, to include subcontractor work and materials.

(REQUIRED REPORT CDRL A001)

1.5.1.1 Production Schedule shall be in Gantt chart format.

1.5.2 Provide a list of all subcontractors with extent of work to be accomplished by each subcontractor. (REQUIRED REPORT CDRL A002)

1.5.3 Provide a list of manufacturer's Technical Representatives to be provided per the particular specification requirement.

1.5.4 Provide a Welder Qualification List, with name, type of qualification(s) and date of last qualification test for each qualification. (REQUIRED REPORT CDRL A003)

1.5.5 Provide manufacturer's product data sheets for all materials to be used by the Contractor to accomplish the requirements of this contract. (REQUIRED REPORT CDRL A004)

1.5.6 Provide Safety Data Sheets (SDS) for all materials to be used by the Contractor to accomplish the requirements of this contract. (REQUIRED REPORT CDRL A005)

1.5.7 Provide a list of key Contractor personnel with names, positions, certifications, telephone numbers, e-mail addresses, and emergency contact information. (REQUIRED REPORT CDRL A006)

1.5.8 Provide a copy of the Contractor's Inspection and Quality Control System with copies of all related forms, reports, and other supporting documentation. (REQUIRED REPORT CDRL A007)

1.5.9 Provide a copy of the Contractor's Environmental Protection and Hazardous Waste Management Plan, including Risk Management/Mitigation plans specific to this contract. (REQUIRED REPORT CDRL A008)

1.5.10 Maintain copies of required federal, state, and local environmental notifications and permits for performance of the work (or a schedule of when they will be obtained).

1.5.11 Each qualification and list shall be available for Government review within 2 hours of request.

1.6 **Progress Meetings:** Weekly progress meetings shall be held at the Contractor's facility near the vessel. The day and time of the meeting shall be set by the SUPERVISOR upon consultation with the Contractor. A review of percentage completion for each work item shall be conducted including the

Contractor's assessment of schedule adherence. Production schedule, corrective action reports, material lead times and other issues shall be discussed.

1.6.1 Review conference with all stakeholders shall be conducted at the availability 25 percent of scheduled Period of Performance. (**Production Start +13 days**)

1.6.1.1 Major discussion points shall include all growth work identified prior up to the 20 percent of scheduled Period of Performance, Milestones, and Schedule.

1.6.2 Review conference with all stakeholders shall be conducted at the availability 50 percent of scheduled Period of Performance. (**Production Start +25 days**)

1.6.2.1 Major discussion points shall include all critical path work, testing work items, Milestones, and Schedule.

1.6.3 Review conference with all stakeholders shall be conducted at the availability 75 percent of scheduled Period of Performance. (**Production Start +38 days**)

1.6.3.1 Major discussion points shall include testing items, Milestones, and Schedule

1.7 **Availability Requirements:**

1.7.1 Arrange for and transport the vessel from the Government facility at Puget Sound Naval Shipyard (PSNS) in Bremerton, WA to the Contractor's repair facility. Upon completion of all work, coordinate delivery of the vessel to the Government facility at PSNS in Bremerton, WA. Both transports of the vessel shall use drawing 600-5283620 as guidance. Provide all materials, labor, and insurance bonds necessary for the safe transport of the vessel to and from the Contractor's facility and for the period that the vessel is located at the Contractor's repair facility.

1.7.1.1 Vessel pickup and delivery shall be scheduled with the SUPERVISOR 5 workdays prior to vessel pickup and delivery.

1.7.2 Provide a separate, lockable, and secure office space for use by the Government On-Site Representative furnished with desk, chair, lights, heat, air

conditioning, electrical outlets, sanitary facilities, Internet service, janitorial service, fire extinguisher, and 24-hour service for the entire contract period.

1.7.2.1 The facility must have a minimum of 240 square feet of floor space.

1.7.2.2 The facility must be located within one-quarter mile of the vessel.

1.7.2.3 Provide a parking area adjacent to the facility. The area must be lighted and accommodate 2 automobiles simultaneously. Designate parking area with sign stating "PSNS PARKING ONLY".

1.7.2.4 Provide QTY (2 EA) keys for all doors and QTY (2 EA) master keys to fit all locks to the SUPERVISOR on the first day the SUPERVISOR is on site.

1.7.2.5 Provide Internet service with QTY (one EA) CAT-5 cable connection or WIFI at 300 MBPS minimum download speed,

1.7.2.6 Desktop Lighting must provide 40-foot candles of illumination measured at the desktop level.

1.7.2.7 Provide QTY (one EA) 15-pound fire extinguisher.

1.7.2.8 Heating and air conditioning must be capable of maintaining the temperature between 65- and 78-degrees Fahrenheit.

1.7.2.9 Provide QTY (one EA) double-pedestal desk, 30-inches wide by 60-inches long.

1.7.2.10 Provide QTY (one EA) swivel chair with adjustable arm rests, and QTY (one EA) straight-back chair.

1.7.2.11 Provide QTY (one EA) 115-volt, shock resistant, double electrical receptacle for each desk. Each receptacle must be a convenient height and located adjacent to each desk.

1.7.2.12 Provide sanitary facilities equipped with drains, hot and cold potable water.

1.7.2.12.1 Hot water must be maintained at 120 to 140 degrees Fahrenheit.

1.7.2.12.2 QTY (one EA): lavatory, water closet, soap dispenser, and towel dispenser.

1.7.2.13 Facilities must be provided 2 days prior to contract start date through 2 days after the contract completion date unless otherwise specified.

1.7.2.14 Provide janitorial services to include sweeping, mopping, and trash pickup on a daily basis.

1.7.2.15 Facility must be in accordance with local building codes, sanitary and current fire regulations. The facility must include smoke alarms and sprinkler systems.

1.7.2.16 Furnishings and equipment in this item must be maintained in a fully operable condition by the Contractor.

1.7.2.17 Facility must be delivered to the Government clean, sanitary, damage free, and vermin free.

1.7.2.18 Provide QTY (one EA) trash can.

1.7.3 Document the result of each inspection requirement throughout this SOW.

1.7.4 Do not operate or utilize existing, newly installed, or repaired shipboard equipment for or in the performance of the Contractor's work effort. Government Representatives will accomplish such operation when required for test and checkout purposes.

1.7.5 Tanks and voids opened shall be clean, dry, and pass Government inspection prior to being closed.

1.7.6 Production Milestone Requirements:

1.7.6.1 Production Start – first day of the Period of Performance.

1.7.6.2 All initial inspections and reports submitted no later than the first 20 percent of the scheduled POP. (**Production Start +10 days**).

1.7.6.3 No later than 3 days prior to the period of performance end date: Turn over 100 percent of spaces to Government. All production complete. (**Production Start +47 days**).

1.7.7 All Government entities shall coordinate with project SUPERVISOR for requirements, site visit and scheduling with the contractor.

1.7.8 Provide photographic evidence of arrival conditions, to include hull, freeboard, weather deck, machinery deck and all tanks/voids. (**REQUIRED REPORT CDRL A009**)

1.7.8.1 Photographs shall be clear, to provide enough visual detail for overall assessment of vessel condition.

1.7.8.1.1 QTY (4 EA) Max photographs per 8" x 11" sheet.

1.7.8.2 Provide QTY (50 EA) photographs upon arrival at contractor discretion to meet the requirements of 1.7.8.

1.7.8.3 Provide QTY (50 EA) photographs at SUPERVISOR direction.

1.8 **Care of Vessel:** The Contractor shall be responsible for the care of vessel equipment, materials, and components related to the work being performed.

1.8.1 Asset shall be maintained in a broom clean condition on a daily basis. Asset shall be free of visual contaminants such as oxide films, oils, grease, dust, dirt, or foreign material.

1.8.1.1 Remove and dispose of industrial debris and extraneous materials from the vessel daily.

(V) "CLEANLINESS INSPECTION"

1.8.1.2 Weekly cleanliness inspection with the SUPERVISOR shall be accomplished, to ensure jobsites are maintained at 1.8.1 level of cleanliness.

1.8.2 Temporary protective measures shall be taken to avoid wear and damage to the Vessel and all associated components. Water shall not be allowed to accumulate in the hull or structure during work performance.

1.8.2.1 Open areas in structure shall be covered to prevent water entry.

1.8.2.2 Install rat guards on each mooring line immediately upon mooring vessel. Each rat guard shall remain installed during 100 percent of moorage time.

1.9 **Workmanship**: Workmanship provided by Contractor employees shall be consistent with the highest standards of commercial marine practices and regulatory bodies. The Government reserves the right to accept or reject material and workmanship nonconformance. Noncompliance with Contractor's quality control processes or rejection of materials/ workmanship by Government will influence past performance rating provided on contract evaluation. Plating and structural member cuts shall be neatly and accurately made with edges cleaned and prepared for welding. Sharp edges shall be ground smooth to avoid injury to personnel. Equipment, machinery, and associated support systems shall not leak oil, water, or other type of fluids. Machinery and equipment shall be protected against weather hazards.

1.9.1 At the time of delivery, work performed shall meet the performance requirements specified. Materials and articles used for work performed shall be of commercial marine quality, conforming to the requirements of the regulatory bodies listed herein, unless specified to be otherwise.

1.9.2 The decision that an area is inaccessible shall be determined by inspection and agreed to by the SUPERVISOR prior to proceeding with the requirements of this SOW.

1.10 **Corrective Action Request (CAR)**: The Contractor is solely responsible for implementation of the Inspection and Quality Control System of 1.5.8. Written responses to contractor generated corrective actions will be provided to SUPERVISOR when requested. Government observer(s) will monitor production work performed at any time where work is ongoing. If, during the performance of work, the Government observer(s) witness work that: fails to meet specifications within this SOW, would lead to an unsatisfactory end product, damages Government property, is a safety violation; the observer will alert the SUPERVISOR. Conditions that fail to meet specifications within this SOW may result in the Contractor being issued a CAR. Contractor must respond

in writing to each issued Method B/C/D CAR within 3 business days unless otherwise specified by the SUPERVISOR. Response must include immediate corrective action taken, preventive action preventive action for recurrence of identified nonconformance, root cause analysis (TABLE ROOT CAUSE ANALYSIS), and Objective Quality Evidence (OQE) for corrective action completed to included estimated completion dates. Contractor must inform the SUPERVISOR when corrective actions are complete for each issued Method A CAR. Response must state that the non-conformance has been corrected”. When a contractor CAR response is unsatisfactory, revised response is required within 3 business days unless otherwise specified by the SUPERVISOR.

1.10.1 Method “A” CAR – Any minor nonconformance that can be corrected/adjusted within 2 workdays of discovery or/and the remediation cost is less than or equal to \$50,000.

1.10.2 Method “B” CAR – Nonconformance where repetitive Method “A” CARs of similar nature and circumstance have failed to result in a satisfactory product or a single nonconformance where corrective action cannot be achieved within three days or remediation cost is greater than \$50,000 but less than or equal to \$500,000.

1.10.3 Method “C” CAR – A systemic nonconformance where multiple Method “B” CARs were issued for similar nonconformities, when a previous Method “B” CAR failed to result in a satisfactory product or a single nonconformance where remediation cost is greater than \$500,000 but less than or equal to \$1,000,000.

1.10.4 Method “D” CAR - When a Method “C” CAR was ineffective in obtaining satisfactory performance or the event was of such a nature that immediate top-level management attention is required to properly address and correct the nonconformance. Factors to consider are costs greater than \$1,000,000, significant impact to scheduled completion, multiple areas of failure within the Quality Management System.

TABLE ROOT CAUSE ANALYSIS

Root Causes	Category Description
- Personnel - Work Practices - Training - Supervision - Procedures - Design - Technical Documentation - Material	- Work Practices- Craftsman knows or understands the requirements, but fails to follow them - Training- Training of employees and subcontractors. - Supervision- Lack of preparation or follow through for the original planned event. - Procedures- Issues with procedures provided by outside agency or another activity utilized during the unplanned event. - Tech Documentation- Issues with DWGs, Specifications or Design aspect. - Material- Failure of material to perform under designed conditions and uses.
Category	Attribute
Work Practices	- Failure to follow Procedure - Use of incorrect or outdated Procedures - Inattention to detail - Improper tools/use of tools
Training	- Nonexistent training or qualification - Content is inadequate - Inadequate training or qualification frequency
Supervision	- Assignment of unqualified personnel - Inadequate direction provided - Inadequate review of worksite or Documents
Procedures	- Procedure contains inadequate or unclear direction - Procedure contains incorrect direction
Technical Documentation	- Error in Drawing or Technical Document - Design deficiency
Material	Material Failure



1.11 Condition Found / Inspection Discrepancy Report (CFR/IDR): Utilize FIGURE A (or Company correspondence containing same information) in transferable media, to communicate to the Contracting Officer via the SUPERVISOR concerning all conditions found, discrepancies noted, and other information requiring formal documentation. Provide the Contractor's recommendation for corrective action to correct the condition found including potential schedule impact. Contractor's estimated pricing (labor-hours and materials) shall be provided separately to the Contracting Officer. Provide all inspection results including detailed measurements, photographs, sketches, and other means to accurately detail inspection results using FIGURE A, with attachments, as necessary. The Government will review and respond to the submitted report. Additional repairs related to the found condition will be tasked if determined necessary by the Government. In the event difficulty is encountered in meeting requirements or difficulty is anticipated in complying with the contract schedule dates, immediately notify the Contracting Officer via verbal communication with the SUPERVISOR. Government receipt of notifications is not to be construed as a waiver of the requirements, delivery schedule, or waiver of rights or remedies provided by law or under this contract, or any other requirements in the SOW relating to jeopardy of contract schedule dates. Reports and notifications do not authorize the Contractor to perform additional work not covered by the contract or authorized by the Contracting Officer. **(REQUIRED REPORT CDRL A010)**

FIGURE A - CONDITION FOUND/INSPECTION DISCREPANCY REPORT

CFR Number _____

Date: _____

From: _____ (Contractor Authorized Representative)

To: PSNS & IMF, NWRMC IND MAN, C/400, Contracting Officer

Via: PSNS & IMF, NWRMC IND MAN, C/101.13, COR

Vessel: Edgecumbe Barge 26' ABS Inspection

Section of Contract Affected: _____

Description:

Impact:

Recommended Action:

C/101.13 Response:

Signature of C/101.13 Representative

Date

Page ____ of ____

1.12 **Interferences:** Contractor pricing offered shall include removal and re-installation of all interferences associated with performance of the required work. The fact that interference is not shown on a plan or specifically identified in the specification item is not justification for a contract change. A physical check of each work site prior to bid submittal is strongly encouraged.

1.13 **Safety Requirements:** Comply with the requirements of 29 CFR 1915, 29 CFR 1910, and 29 CFR 1926, state and local laws, rules, and ordinances during contract performance.

1.13.1 **General:** Contractor shall be responsible for the compliance of the above requirements by all subcontractor personnel performing work related to this contract. In cases of conflict between the above regulations, the stricter of the requirements shall take precedence.

1.13.2 **Accidents and Injuries:** Immediately report injuries to Contractor and subcontractor employees to the SUPERVISOR. Investigation shall be completed within 7 workdays of each accident/injury. Document a report that includes narratives of the circumstances leading up to and including the injury and any actions (immediate and contemplated) taken to mitigate future reoccurrences of injuries. **(REQUIRED REPORT CDRL A011)**

1.13.3 **Scaffolding:** Scaffolds and ladders shall meet the requirements of 29 CFR 1915.71 and 29 CFR 1915.72.

1.13.3.1 Install a swinging gate guardrail at each scaffold ladder and through-platform access opening.

1.13.3.2 Install standard railing around the non-access sides and a swinging gate guardrail on the entrance side of through-platform access openings.

1.13.3.3 Install standard toe-boards around all scaffold floor openings, except at the entrance to the swinging gate guardrail, on all exposed sides of through-platform access openings.

1.13.3.4 Develop detailed engineered drawings and specifications for scaffolds when OSHA, ANSI, or manufacturers' designed guidelines do not exist, such as for "hanging scaffolding". Drawings and specifications shall be designed by a registered professional engineer and meet the following:

1.13.3.4.1 Maintain on the jobsite a copy of the detailed engineering drawings and specifications showing the sizes and spacing of members and the calculations developed for the engineered design(s).

1.13.3.4.2 Construct, load, and maintain scaffolding in accordance with the registered professional engineer's design.

1.13.3.5 Install a commercially available scaffold tag system on all scaffolds. The tagging system, as a minimum, shall be prominently attached to the scaffold structure, consist of green colored "Okay to Use" tag(s), red colored "Danger" tag(s), and accomplish the following:

1.13.3.5.1 Apply all tags on the scaffold structure as close as practicable to each ladder access and/or entry point.

1.13.3.5.2 Apply a red tag to indicate to users that the scaffold is being dismantled, is not yet completely erected, or is unsafe for use.

1.13.3.5.3 Apply a green tag to indicate to users that the scaffold is safe for use, is compliant with all OSHA regulations, and other applicable requirements.

1.13.3.5.4 Ensure green tags contain the following information as a minimum: the location of the scaffolding structure; a unique reference number to identify each separate scaffolding structure, the date the scaffolding structure was first erected; the name(s) of the scaffold builder(s); the load rating of the scaffold structure; and the Competent Person's printed name and signature.

1.13.3.5.5 Green tags shall be checked and initialed daily by the Contractor's scaffolding competent person.

1.13.3.6 Scaffolding shall be free standing and self-supporting, free from any attachment to any lifeline, handrail, antenna, safety equipment, electronic fixture, or vessel equipment.

1.13.3.7 Structures and Openings shall meet the requirements of 29 CFR 1915.73.

1.14 **Hot Work/Welding:** Prior to performance of any hot work on the vessel, certify a safe atmosphere exists in and around the work area before starting any

work that may produce the heat of ignition, sparks, or flame. Accomplish steps to make each area safe for hot work. A copy of the gas free certification shall be posted in accordance with applicable OSHA requirements.

1.14.1 All welders shall be certified in accordance with American Weld Society (AWS) and American Bureau of Shipping (ABS).

1.14.1.1 Perform non-destructive testing of new welds, when invoked by the requirements of section 3.0 of this SOW, in accordance with the requirements of (ABS).

1.14.2 All welding and Non-destructive testing; to include weld filler materials, plating materials, weld designs, weld repairs, and equipment shall be in accordance with ABS Rules for Materials and Welding 2020 Part 2.

1.14.3 ABS Surveyor provide and accomplish inspections of each weld process, weld joint preparation, fit up, root pass, back gouge, final cover pass, and Non-destructive Test and Inspection, as required for barge certification.

1.14.3.1 ABS shall have access to the barge during various stages of progress, for example the hull access cut and repair inspections, as required.

1.14.3.2 Document and submit 100 percent complete ABS inspections. (REQUIRED REPORT CDRL A032)

1.14.3.3 Each weld quality requirement, process requirement, and inspection shall be accepted by the ABS surveyor. SOW work requirements will not be considered complete until 100 percent of inspections are documented as satisfactory by the ABS surveyor.

1.14.4 Weld defects shall be repaired in accordance with ABS requirements.

1.14.5 Remove existing preservation coatings prior to each expected welding, cutting, and hot work operation.

1.14.6 Preservation/coatings shall be removed 4 inches in each direction and each opposite side.

1.14.7 Welding shall be sequenced to keep the thermal expansion of structures to a minimum.

1.14.8 Minimum preheat accomplish 100 percent of each weld joint to minimum 70 degrees Fahrenheit prior to each weld application.

1.15 **Threaded Fasteners:** Except as otherwise specified herein, or on the drawings listed, threaded fasteners shall be in accordance with applicable ASTM standards. Coarse thread series shall be used unless the component's design indicates a necessity to use fine thread series. Fastener clearance holes shall be 1/16-inch oversize unless specified in the requirements of 2.0. Black oxide coated brass and cadmium plated fasteners are prohibited. Where drawings specify the use of brass fasteners on a fuel or fuel oil tank, Corrosion Resistant Steel (CRES) 316 fasteners shall be substituted in lieu of brass. Consult the requirements of MIL-STD-777 and Naval Ships' Technical Manual (NSTM) Chapter 075 for guidance in instances where there is a question as to the material to be utilized.

1.15.1 Threads on externally threaded fasteners, after being installed and tightened, shall protrude a minimum of 2 threads to a maximum of 5 threads beyond the face of the nut.

1.15.1.1 Minimum thread protrusion of self-locking (plastic insert) nut installations shall be flush with the face of each nut.

1.15.1.2 Self-locking fasteners shall not be re-used.

1.15.2 Clean, chase, and tap exposed threaded areas of each fastener and threaded socket.

1.15.3 Tighten threaded fasteners in the sequence, patterns, and the torque values, when not otherwise specified, using the requirements of Section 4, paragraph 075-4.1 through 075-4.6.2 of NSTM Chapter 075 for guidance.

1.16 **Preservation:**

1.16.1 Measure, record, and maintain Objective Quality Evidence (OQE) information on Appendices one through 8 throughout the entire performance of preservation operations. Documenting OQE information for the in-process shall be made immediately available for onsite review upon request by the SUPERVISOR. Paint qualifications must meet the requirements of NBPI, NACE or SSPC. (REQUIRED REPORT CDRL A012)

1.16.1.1 Appendices one through 8 shall be in accordance with Reference QA_Appendix_Forms_C101_2023_Contractors.pdf

1.16.1.1.1 All spaces that are not applicable shall be marked N/A.

1.16.1.1.2 Unused sections shall be crossed out and marked N/A.

1.16.1.1.3 Any errors shall be one lined through; initialed and dated by the person making the correction.

1.16.1.1.4 Any markings that are illegible or unclear shall be considered invalid.

1.16.2 Comply with the coating manufacturer's requirements for the establishment and maintenance of environmental conditions and for coating thickness applications throughout the entire removal and preservation operation(s).

1.16.2.1 Calibrated Electronic Data Logger record and maintain environmental conditions OQE during the entire preservation operations. Data shall be made immediately available for onsite review upon request by the SUPERVISOR. (REQUIRED REPORT CDRL A013)

1.16.3 Apply contrasting colors between the primer, stripe, intermediate, and topcoat using coating products that are from the same manufacturer and compatible with each other; other system preservation coatings shall be different in color from both the preservation coating over which it is being applied and the next coating in the system; no two coatings can be similar in color and shall be applied in accordance with the manufacturer's product data sheet.

1.16.4 Apply stripe coat to all edges, weld seams, welds of attachments and appendages, cutouts, corners, butts, foot/handholds, including back side of piping, underside of beams and other mounting hardware (non-flat surface) after each previous full coat has dried. The stripe coat shall encompass and extend a minimum of one inch beyond the intersection of two flat surface planes, each radii edge, and all perpendicular shapes, edges, or angles.

1.16.5 Solvent clean surface prior to the start of each blasting, and preservation operation. Accomplish the requirements of Surface Preparation Specification SSPC-SP-1 of Systems and Specifications, SSPC Painting Manual, Volume 2.

1.16.6 Accomplish touch-up preservation for new and disturbed surfaces shall match surrounding coating system(s) in accordance with the manufacturer's product data sheets.

1.16.7 Feather edges of well adhered paint remaining after cleaning for all surface preparation methods. Visible areas of defective existing paint shall be removed until an area of completely intact and adhering paint is attained around the defective area by feathering (tapering) the edges of tightly adhering existing paint at an approximate 30-degree slope into the newly prepared bare metal surface thus preventing application of new paint over loose or cracked paint.

1.16.8 Paints shall not be thinned.

1.16.9 Hull and freeboard preservation coatings shall be fully cured prior to undocking and/or shifting the vessel. All other preservation coatings shall be fully cured prior to final delivery to the Government.

1.16.9.1 Preservation coatings are considered fully cured when they meet the requirements of the applicable manufacturer's product data sheets minimum cure to service time schedule and are dry throughout their thicknesses.

1.16.10 Conduct Surface Chloride or Surface Conductivity measurements.

1.16.10.1 Surface Chloride; shall not exceed 3 ug/cm² (30 mg/m²) for immersed or 5 ug/cm² (50 mg/m²) non-immersed applications. Surface Conductivity: shall not exceed 30 micro-siemens/cm immersed or 70 micro-siemens/cm non-immersed applications.

1.16.10.2 Readings shall be documented on Appendices 4.

1.17 Environmental and Hazardous Waste Management: Comply with all federal, state, and local laws, codes, regulations, acts, ordinances, and rules for Environmental Protection and Hazardous Waste Management / Disposal, including obtaining required licenses, generator identification number, and permits. Designate one employee responsible for Environmental and Hazardous Waste Management. All contract, manifests, invoices, and other documents related to the removal, storage, handling, transportation, or disposal of hazardous waste shall bear a generator identification number issued to the Contractor pursuant to applicable law. Take actions to mitigate creation of hazardous waste by recycling and other available methods. All hazardous waste generated by

work under this contract is anticipated to be Contractor generated and shall be the responsibility of the Contractor. Estimated hazardous waste associated with work required in these Specifications is listed below:

<u>Product</u>	<u>Amount</u>
Cleaning Solvents	0 gallons
Paint (enamel, latex, epoxy, thinners, oil based, rubber, non-skid, removers)	0 gallons
Blast Media and paint chips contaminated with heavy metals and other contaminants	0 pounds
Tank Solids	50 Pounds
Tank Liquids	1,000 Gallons

Hazardous waste generated during the actual performance of the work may vary in type or amount from the waste listed above. The above is an estimate of the amount of waste anticipated to be generated; not an estimate of the amount of work in generating that waste. **(REQUIRED REPORT CDRL A014)**

1.17.1 The Contractor shall ensure that their cost proposal for the work proposed herein includes reasonable compensation for all anticipated cost of managing and disposing of the wastes listed above, and/or created or generated in the execution of the contract issued, in accordance with the requirements of 42 USC 6901-6992 Rev 1996, 49 USC 5103, 10 USC 7311 Rev 1/22/11. The Contractor shall include a 5-way break down of the anticipated costs associated with this effort in their offer.

1.17.2 Consider coatings to contain heavy metals (e.g., lead, chromium, cadmium), hexavalent chromium, crystalline silica, and/or other toxic or hazardous substances until laboratory analysis establishes otherwise. Comply with all procedures and precautions required for employee protection, sampling, removal, and disposal.

1.17.3 Consider insulation, lagging, and mastics to be asbestos-containing material (ACM) until it can be established by laboratory analysis, or other reliable method(s), that the material does not contain asbestos. Ensure compliance with the requirements of 29 CFR 1915.1001.

1.17.4 Consider materials, debris, and/or residue associated with ventilation systems to contain toxic/hazardous substances including but not limited to cadmium, chromium, and PCBs until laboratory analysis establishes otherwise.

1.18 **Materials:** All Contractor furnished materials (CFM) installed shall be new. Materials shall be as indicated on the listed drawings, publications or as specified in this Specification. The use of asbestos, polychlorinated biphenyl (PCB), magnesium, cadmium, or mercury in any materials is prohibited.

1.18.1 Procure/Order 100 percent of materials within 5 workdays of contract award, to support the production schedule and delivery of the vessel within in the contracted period of performance.

1.18.2 Status Report Document 100 percent of CFM that is required to accomplish the requirements of this SOW. (REQUIRED REPORT CDRL A015)

1.18.3 Provide copy of all material certifications providing a minimum of manufacturer, ABS Certification, heat number, batch number, and manufacturing date. (REQUIRED REPORT CDRL A016)

1.19 **Weight Control:** Institute a weight control program in which actual weights, for removed, relocated, and installed equipment/structure, are measured by either weighing or from manufacturer certified data. The location of the center of the load must be recorded along with each weight entry in the weight control report. This location shall show the distance vertically from the baseline of the vessel, longitudinally from amidships and tangentially from the centerline of the vessel. Items that are permanently removed and replaced with the identical item, in the same location, are not required to be included in the weight control program. Items that are replaced with like items of differing material quantity or new installations shall be included in the weight control program.

1.20 **Label Plates:** Label plates installed in locations exposed to the weather shall be CRES and shall be prepared by the “Metal-Photo” process and sealed to exclude moisture between the plate and the mounting surface. Label plates shall be installed to assure maximum visibility. Label plates shall be attached with CRES screws or adhesive, using drawing S2803-980209 Rev L for guidance. Where adhesive is used, surface preparation shall be in strict accordance with the adhesive manufacturer’s instructions. Test label plates shall be attached on or adjacent to all equipment or components tested without interference to equipment

or component operation. The label shall identify the equipment or component, the test load, the dynamic and static load (if applicable), the rated load, the date of testing, and the activity performing the test, using the layout format of drawing S2803-980209 Rev L for guidance.

1.21 Integrated Logistic Support: Maintain legible original copies in approved transferrable media, of the following original equipment manufacturers (OEM) information: operating instructions, manuals, drawings, catalog cut sheets, warranty information, and all other supporting documents, for all new equipment, parts, and supplies installed.

1.22 Gas Free: Provide certified Marine Chemist, Certified Industrial Hygienist services, and Shipyard Competent Person services as required by 29 CFR 1915 for Contractor and Government Personnel.

1.22.1 Consolidate and provide the Contractor's current confined space entry/gas free certification processes into a single and comprehensive document as part of a confined space/gas free certification program. **(REQUIRED REPORT CDRL A017)**

1.22.1.1 If rescue plan utilizes a third party, provide documentation of written agreements from third party. **(REQUIRED REPORT CDRL A018)**

1.22.2 Provide a list of Contractor's National Fire Protection Association (NFPA) Marine Chemist certifying gas-free conditions for this project and a list identifying Contractor employees designated as "Competent Persons" in accordance with 29 CFR 1915.7. **(REQUIRED REPORT CDRL A019)**

1.22.3 Post, update, and maintain gas free certificates in accordance with 29 CFR 1915 during the entire contract period of performance.

1.23 General Requirements for Testing: Test all Contractor affected equipment, systems, and subsystems for compliance with this SOW and commercial marine standards and practices. Demonstrate workmanship, alignment, strength, rigidity, water tightness, functionality, and suitability for the purpose intended.

1.23.1 Develop a Test and Inspection Plan for each Test and Inspection required by zero and first tier references, each symbol (I) (V) (Q) (G)

test/inspection and by this SOW, containing paragraph number or listed applicable reference requiring Tests and/or Inspection; identification of each component/system; description of each Test and/or Inspection; accept/reject criteria; date scheduled; Government notifications; date completed; and each Test and/or Inspection result. Tests and inspections the Contractor deems necessary to substantiate product conformance shall be included. (REQUIRED REPORT CDRL A020)

1.23.1.1 Fully integrate the Test and Inspection Plan with the Contractor's production schedule.

1.23.1.2 Provide the number of tests and inspections for a given system, component, or installation required to demonstrate required satisfactory performance. This section specifies tests to be performed in support of the work performed.

1.23.2 Provide a detailed description of preoperational checks to be conducted, operations to be performed, and the parameters to be satisfied. Develop data sheets which provide spaces for recording the quantitative values measured during the test, for each test procedure. Specify the minimum and maximum allowable tolerance limits for each measured value. Develop signature sheets for fill-in by the person conducting the test, the Government witness present, and the date of the test. Block diagrams, simplified schematics or diagrams may be used to clarify the procedure or test method. Include a comment sheet to record significant events and observations by either the Government witness or Contractor.

1.23.3 Provide materials, supplies, utilities, equipment, calibrated equipment/ instrumentation, services, and personnel to accomplish each test. Incorporate safety precautions into each test procedure. Prepare all test documentation during the development and implementation of the acceptance test.

1.23.3.1 Whenever Government support is required for the operation of vessel equipment and/or systems, a minimum of 3 workdays prior notification of the operational/test support requirement shall be provided to the SUPERVISOR.

1.23.3.2 Tests/inspections and equipment/systems operation requiring Government operational support shall be scheduled for accomplishment during normal day shift working hours, as much as practicable.

1.23.3.3 Provide a minimum of 3 workdays prior notification to the SUPERVISOR for tests/inspections and equipment/system operations requiring Government operational support for work being performed on weekends, after normal dayshift working hours, and immediately before and/or after Federal holidays.

1.23.3.4 Notify the SUPERVISOR a minimum of 3 workdays prior to each test/inspection. Equipment shall not be adjusted during acceptance testing. The SUPERVISOR may countersign the test/inspection report; however, the Government's signature does not constitute approval or concurrence in any aspect of the testing/inspection, other than the presence of the Government at the test.

1.23.4 Perform tests/inspections using the applicable test procedure for systems and subsystems. Demonstrate compliance with the Specification requirements by satisfactory completion of the test. Complete all work prior to initial testing of that system or subsystem and in a timeframe sufficient to allow for correction of deficiencies prior to dock trials, sea trials, or other applicable milestones established in these specifications. Perform adjustments and testing prior to each scheduled acceptance test.

1.23.4.1 Gage range shall be such that test pressure is in the middle third of each scale.

1.23.4.2 Stop the test/inspections if the equipment, system, or subsystem tested fails to satisfy the acceptance criteria of the established test procedure. Identify the cause and correct.

1.23.5 Accomplish (G)-Point (Government Notification):

1.23.5.1 (G) is a symbol inserted in a SOW paragraph to establish a point in the sequence of accomplishment of work at which time the SUPERVISOR must be notified by the prime Contractor in all cases to permit observation of a specific test or inspection (I) (V) by the government. When the symbol (G) precedes tests or inspections in a SOW paragraph which are applicable to more than one action, the symbol (G) identifies the action required, e.g., (G) "HYDROSTATIC TEST". When more than one unit is involved, the (G) notification requirement applies to each unit.

1.23.5.2 For tests and inspections involving (G)-points, records must be documented upon acceptance or rejection and electronic copy provided to the

SUPERVISOR within 24 hours of conclusion of each (G)-Point. **(REQUIRED REPORT CDRL A021)**

1.23.5.3 Government will consider the work in accordance with SOW paragraph incomplete if the Contractor's documentation and records are not complete.

1.23.5.4 Proceed with the test or inspection if the Government Representative is not present, provided the required advance notice has been furnished to the SUPERVISOR, and the Contractor has completed and documented the preceding tests and inspections.

1.23.5.5 A partial test or inspection requiring (G) point notification may be accomplished in the event that all work cannot be completed, and work progress would be delayed in waiting for total completion of work. Comply with the requirements of 1.23.6 when the incomplete work is completed and ready for the remainder of the test or inspection. Note partial inspections on the test or inspection form.

1.23.5.6 A qualified Contractor representative must be present to accomplish, accept or reject and document tests or inspections associated with the symbol (G).

1.23.6 Accomplish (I) (V) and (Q) Checkpoints:

1.23.6.1 Accomplish (I), (V) and (Q) tests/inspections that do not have associated (G)-points, with qualified and/or currently certified personnel where required by technical documents (e.g., NBPI, NACE, nondestructive testing, electrical cableway inspection, etc.)

1.23.6.2 (I) inspections require verification and documentation by a separate individual, other than the person who has accomplished the work, who is qualified as an inspector.

1.23.6.3 (V) inspections require verification and documentation by the qualified tradesperson, trade supervisor, or inspector.

1.23.6.4 (Q) inspections require verification and documentation by a qualified Technical Representative in accordance with 1.26.

1.23.6.5 The authority to accomplish, document, accept and reject (I) and (V) inspections may be delegated to qualified subcontractor personnel, without regard to geographical location, subject to SUPERVISOR approval.

1.23.6.6 The authority to witness or perform, document and accept/reject (I)(G), (Q)(G), and (V)(G) tests and inspections is a prime Contractors responsibility but, subject to SUPERVISOR approval within a 50-mile radius of the Contractors plant nearest to the place of performance of the contract, may be delegated to subcontractors who are MSRA or ABR agreement holders, SSPC QP1 certified, NDT certified, or have a current QMS accepted by the SUPERVISOR.

1.23.6.7 The Contractor may delegate responsibility to subcontractors to perform, document and accept/reject (I)(G) and (V)(G) tests and inspections performed at plants located outside a 50-mile radius of the Contractor's plant nearest to the place of performance of the contract subject to SUPERVISOR prior approval.

1.23.6.7.1 For work being performed outside a 50-mile radius of the place of contract performance, the prime Contractor must submit via approved transferrable media, of purchase orders to the SUPERVISOR within 2 workdays or otherwise as directed by the Government, prior to issue of purchase order and shipment of equipment. For Contractors who do not utilize purchase orders as a vehicle for accomplishing work within their company, a report identifying the delineation of the specific SOW requirements, in lieu of the purchase order must be submitted to the SUPERVISOR.

1.23.7 Document the result of each test requirement throughout this SOW.

1.24 **Process Control Procedure (PCP) Requirements:** Not Used.

1.25 **Compartment Arrival/Closeout Inspection:** Not Used.

1.26 **Technical Representative:** Not Used.

1.27 **Heavy Weather Plan:** Provide a written plan that must be implemented during gales, storms, hurricanes, and destructive weather, including mooring calculations in accordance with 845-6686999 Rev D Vessel Water Depth, Mooring and Hull/Appendage/Clearance Requirements for Transit and Berthing, Calculations for Mooring Systems, using Mooring and Towing and UFC 4-159-

03, Mooring Design for guidance. The documented Heavy Weather Plan must be submitted for a document review and acceptance. The Contractor must have an acceptable documented Heavy Weather Plan in place no later than 15 days prior to availability start date. The Heavy Weather Plan must be available for periodic conformity audits throughout the contract. **(REQUIRED REPORT CDRL A022)**

1.27.1 Ensure the plan designates responsibility and implements procedures for prevention of damage to the vessel. This includes periods when the vessel is physically located in private Contractors' plants; during times when work on the vessel requires openings to hulls or decks; and when Contractor owned/furnished floating equipment is tied alongside the vessel. The plan must contain specific responsibilities and detailed actions to be taken during the weather conditions listed below.

1.27.1.1 Conditions where there is substantial advance warning for approaching adverse weather are addressed by the following 4 categories:

1.27.1.1.1 Gale/Storm/Hurricane Condition IV: Trend indicates a possible threat of destructive winds of force indicated within 72 hours.

1.27.1.1.2 Gale/Storm/Hurricane Condition III: Destructive winds of force indicated are possible within 48 hours.

1.27.1.1.3 Gale/Storm/Hurricane Condition II: Destructive winds of force indicated are anticipated within 24 hours.

1.27.1.1.4 Gale/Storm/Hurricane Condition I: Destructive winds of force indicated are anticipated within 12 hours or less.

1.27.1.2 Conditions where there is little or no advance warning for approaching adverse weather are addressed by the following 2 categories:

1.27.1.2.1 Thunderstorm/Tornado Condition II: Destructive winds accompanying the phenomenon indicated are reported or expected in the general area within 6 hours. Lightning and thunder are also anticipated.

1.27.1.2.2 Thunderstorm/Tornado Condition I: Destructive winds accompanying the phenomenon are imminent. Lightning and thunder are also anticipated.

1.27.1.3 Ensure that the plan contains, as a minimum, the following information as dictated by conditions listed in 1.27.1:

1.27.1.3.1 Steps to be taken to remove or secure staging items or equipment on decks of the vessel, pier or dry dock, including cranes that could become windborne.

1.27.1.3.2 Protection of the vessel from damage from other floating equipment, such as barges, doughnuts, work floats, and other ships, craft, barges, and lighters.

1.27.1.3.3 Provisions for protection of government equipment and material in custody of the Contractor from damage by pier side flooding.

1.24.1.3.4 Provisions for removal of temporary hoses, welding lines, air lines, oxygen/acetylene lines, etc., extending through watertight closures.

1.27.1.3.5 Provisions for security, emergency fire and flooding protection, emergency shipboard dewatering and fire main capability, emergency shipboard electrical generation, and emergency shipboard communications.

1.27.1.3.6 One portable dewatering pump and associated equipment for each 100 feet of ship's length in addition to the dewatering equipment provided by 845-6686999 Rev D Vessel Water Depth, Mooring and Hull/Appendage/Clearance Requirements for Transit and Berthing must be available on the damage control or main deck adjacent to damage control lockers or temporary damage control equipment boxes. Pumps must be capable of providing a minimum of 200 gal/min at a discharge head of 50 feet of dewatering capacity and can be used at the scene of a casualty within 3 minutes of receiving an alarm. Additional dewatering capacity to provide 1,000 gal/min at a discharge head of 50 feet at the scene must be available within 15 minutes. During the waterborne overhaul period, no damage control system associated with flooding prevention and control, or any portion thereof must be removed or made inoperable without prior notification of the SUPERVISOR and to the casualty-control station and until a back-up system has been established.

1.27.1.4 Ensure provisions for access to the ship for personnel and emergency equipment during and immediately following the storm consistent with prudent safety precautions. Assurance that all hull/deck openings are made

watertight. Steps to be taken to secure floating piers during high winds/high tides.

1.27.1.5 The name and telephone number (business and residential) of the Contractor's single point of contact. This person must have the authority to commit the Contractor to take necessary actions as requested by the SUPERVISOR.

1.27.2 Ensure the plan contains the following mooring related information:

1.27.2.1 Specify steps to be taken to secure the vessel to Contractor's pier, dry dock, graving dock, marine railway, or Contractor's other facility. Information must define specific precautions to be taken and supporting calculations, to include limits of docking blocks and dock stability for both normal and heavy weather conditions. Calculations for heavy weather configurations must include wind and tidal considerations.

1.27.2.2 Provide the heavy weather state at which the ship must be undocked.

1.27.2.3 Submit mooring calculations for the worst anticipated loading condition during the availability. For vessels with a self-compensating fuel system, the loading condition must show the self-compensation fuel system full of water, fuel, or some combination of fuel and water, projecting the worse possible condition as shown in calculations for maintaining ship's stability. Calculations may require re-submittal if significant changes occur from the original estimate on which the calculations were based.

1.27.2.4 For ships in dry dock, provide limits and supporting calculations for listed conditions. Analyze both the "normal" dock configuration and the "heavy weather" configuration.

1.27.2.4.1 Maximum safe wind speed and surge for side block strength and stability. Include maximum loading of the side blocks on the vessel.

1.27.2.4.2 Maximum safe wind speed and surge for dry dock strength and stability.

1.27.2.4.3 Surge required to float the vessel.

1.27.2.4.4 Table or graph showing safe combinations of wind speed and surge.

1.27.2.5 For ships pierside, provide limits and supporting calculations for vessel loading conditions specified in 1.27.2.3. Analyze the "heavy weather" mooring configuration that would be used during the conditions specified in 1.27.1. Analyze worst-case wind directions including frontal, broadside, and quartering.

1.27.2.5.1 Maximum safe wind speed for mooring strength. Include strength of pier, pier fittings, mooring lines, and shipboard fittings. Maximum applied load on any mooring line must be the breaking strength of the mooring line divided by 2.5 (factor of safety of 2.5).

1.27.2.5.2 Maximum safe surge for mooring.

1.27.2.5.3 Maximum safe elongation of mooring lines. Include the following information: Size and type of mooring line; Percent elongation of mooring line at failure; Tattletale-free length and length between attachments.

1.27.2.5.4 Sketch, showing size, type, and location (vertical and horizontal angles) of all securing devices including fenders, bumpers, and camels.

1.27.2.6 Include the following statement, providing the necessary data: Vessel _____ can be safely moored to withstand a maximum of ____ mph winds with a ____ knot current and a ____ foot storm surge.

1.28 Foreign Material Exclusion (FME): Install and maintain temporary expandable FME plugs, blanks, or other hard and durable protective covers, that are painted blaze orange to maintain system cleanliness from environmental dangers.

1.28.1 Install and maintain blanks/plugs, studs, nuts and bolts, painted blaze orange for use foreign material exclusion (FME) immediately upon openings in equipment, valves, and piping systems not subject to pressure to prevent entry of foreign material and protect flanges and threaded areas.

1.28.1.1 Existing system fasteners used for blanking that will be used for installation are excluded from the requirement for blaze orange color.

1.28.1.2 FME may be used for systems normally under pressure but are tagged-out for maintenance.

1.28.1.3 Wood plugs, wood, or wood products are prohibited for use in tanks/voids, and as blanks on pressurized systems.

1.28.1.4 Wood products, including plugs, are permitted for use as FME external to the vessel for hull penetrations, and non-pressurized systems to include gravity drain piping.

1.28.1.5 The use of cloth, polyvinyl sheet, paper, tape and rubber sheeting, as FME prevention devices is prohibited.

1.28.2 FME shall be install in hull openings and surfaces to prevent system contamination or equipment damage from abrasive blasting, surface preparation, and preservation operations.

1.28.3 Establish an FME log, record, and identify the location of each protective covering installed and removed, including each corresponding and specific measurement(s), location(s), and/or area(s) in relationship to structural members or from suitable hull reference points. **(REQUIRED REPORT CDRL A023)**

1.28.3.1 Remove each temporary FME plug, blank, and protective covering, record removal date in FME log, and restore each system to its original configuration upon final completion of testing, blasting, and preservation operations.

2.0 REFERENCES: All references shall be of the latest issue and/or revision or as indicated in the Consolidated List of References. It is incumbent on the Contractor to maintain current commercial specifications and be thoroughly familiar with laws, regulations, and ordinances. Contractor is prohibited from using Government provided technical information for any purpose other than that contemplated by the specifications. Contractor is prohibited from further use, release, or disclosure of this information. References provided on SAM.gov, safe.apps.mil and/or CD-ROM, including any information contained thereon, shall be destroyed in accordance with the instructions as stated below, at completion of the contract. After award, for those Contractors in receipt of reference material from SAM.gov, safe.apps.mil and/or CD-ROM and not awarded a contract shall destroy all information received in accordance with the instructions stated below. This applies to all physical and electronic copies of reference material associated with this contract award.

DESTRUCTION NOTICE: Destroy by any method that will prevent disclosure of the contents or reconstruction of the documents.

2.1 Order of Precedence: Unless otherwise noted within this specification, hierarchy, priority, or order of precedence of requirements shall be as follows:

- 2.1.1 Specification Definite Item
- 2.1.2 Specification General Requirement
- 2.1.3 Military/Navy Drawings
- 2.1.4 Military/Navy Technical Manuals
- 2.1.5 Commercial Drawings
- 2.1.6 Commercial Technical Manuals
- 2.1.7 Commercial or Industrial Standards
- 2.1.8 Commercial Practices

2.2 Discrepancies: Drawing, technical reference, or other discrepancies associated with this specification shall be reported using the Request for Specification Clarification form, during proposal, or the “Condition Found / Inspection Discrepancy Report” form (FIGURE A) during contract period of performance.

CONSOLIDATED LIST OF REFERENCES		
The following publication and drawing requirements are applicable to this SOW:		
Publication Number		Title
1	29 CFR 1910	Occupational Safety and Health Standards
2	10 USC 7311	Repair or Maintenance of Naval Vessels: Handling of Hazardous Waste

3	29 CFR 1915	Occupational Safety and Health for Shipyard Employment
4	29 CFR 1926	Safety and Health Regulations for Construction
5	42 USC 6901 to 6992	Resource Conservation and Recovery Act of 1976
6	49 USC 5103	Hazardous Materials Transportation Authorization Act of 1994
7	ANSI/AWS D1.1	Structural Welding Code, Steel
8	MIL-STD-777	Schedule of Piping, Valves, Fittings and Associated Piping Components for Naval Surface Ships
9	NSTM Chapter 074 Volume One	Welding and Allied Processes
10	NSTM Chapter 075	Threaded Fasteners
11	UFC 4-159-03	Mooring Design
12	SSPC Painting Manual, Volume 2	Systems and Specifications
13	QA Appendix Forms C101 2023	Contractors.pdf
14	NAVSEA 9997/1 through 9997/5 Final Docking Report Forms (2025)	
Drawing Number		Title
1	100-5283618-A	Edgecumbe Barge Structural Modifications – Mid Body
2	100-5283619-A	Edgecumbe Barge Removals & Modifications – Bow & Stern
3	100-7281566 Rev D	RC Barge Paint Sched and Zinc Anode
4	114-5322269-B	Edgecumbe WT Bulkheads Required for Stability
5	145-6773239	Edgecumbe Barge Compartmentation Arrangement
6	600-5283620	Primary Towing Equipment Drawing
7	803-921865 Rev T	Cathodic Protection
8	845-7281575	Edgecumbe Barge Docking Drawing
9	845-6686999 Rev D	Vessel Water Depth, Mooring and Hull/Appendage/Clearance Requirements for Transit and Berthing
10	No Dwg Number	Edgecumbe Deck Arrangements
11	S2803-980209 Rev L	Bureau of Ships, Label Plates
12	S2804-921819 Rev C	Distinguishing Figures

3.0 **WORK REQUIREMENTS:**

3.1 **Docking And Undocking The Vessel:**

3.1.1 Personnel and equipment: Furnish qualified and experienced personnel, facilities, and the necessary equipment to safely dock and undock the vessel, including:

3.1.1.1 A Dock Master currently qualified and certified by the Contractor and accepted by the Government Docking Observer (GDO).

3.1.1.2 Supporting employees to accomplish the docking and undocking evolutions.

3.1.1.3 Tugs, pilots, and divers required for the complete transfer, docking, and undocking of each vessel. At the docking conference, the GDO shall review and approve the number and size of the tugs to be used.

3.1.2 Docking Procedure: Develop and provide a docking procedure prior to dry docking, which shall include the following: (REQUIRED REPORT CDRL A024)

3.1.2.1 Operating practices.

3.1.2.2 Safety and security plans.

3.1.2.3 Special precautions or actions required by the characteristics of the dock, marine railway, or other docking methods.

3.1.2.4 Pumping/flooding schedule.

3.1.2.5 Specific list, trim, and stability requirements.

3.1.3 Blocking Plan: Provide a blocking plan to set and align the docking blocks, using drawing 845-7281575 for guidance. (REQUIRED REPORT CDRL A025)

3.1.3.1 Develop a docking plan in the form of a scaled drawing including: block size, block spacing, and location of vessel with respect to the blocks at landing. Position blocking to ensure that equipment on the surface of or protruding from the hull shall not be damaged.

3.1.3.2 Determine the docking clearances utilizing references for guidance and the following requirements: The minimum clearance between the side of the vessel (including fixed projections) and the permanent dock structure (such as wingwalls, crane rails, piers, etc.) shall be 12 inches during vessel movement into or out of dock and allow for safe work practices accomplished inside the dock. Minimum vertical clearance between the keel of the vessel and the dock floor shall allow for safe accomplishment of work and provide sufficient access for all inspections. Hull openings shall not be obstructed.

3.1.3.2.1 Minimum vertical clearance between the keel of the vessel and the dock floor shall be 42 inches for barges, excluding deep draft barges.

3.1.3.3 The blocking plan shall be accompanied by a written request for approval to the GDO a minimum of 5 workdays prior to the scheduled docking date. The use of haul, fixed, or fitted blocks shall be clearly stated.

3.1.3.4 The use of universal docking blocks is prohibited.

3.1.3.5 Maintain supporting data substantiating that:

3.1.3.5.1 The block loading pressures are limited to a maximum of 5.4 long tons per square foot.

3.1.3.5.2 The number, distance from centerline, and square footage of the side blocks per side is sufficient to resist overturning moments.

3.1.3.5.3 At least 90 percent of blocks land on longitudinal strength members or main transverse bulkheads.

3.1.3.6 Blocking shall be constructed of hardwood or composite hardwood/concrete and soft caps, with at least 2 inches of soft cap thickness.

3.1.3.6.1 The term “hardwood,” for the purposes of this contract, includes White Oak, California Laurel, Oregon Myrtle, Iron Wood, Blue Gum, American Rock Elm, or preserved Red Oak. The normal lifespan for hardwood blocking is approximately 10 years.

3.1.3.6.2 Woods acceptable for use as soft caps are Douglas Fir, Tamarack, Long Leaf Pine, or Hemlock.

3.1.3.6.3 Composite blocking showing evidence of spalling and cracks, or chipped and damaged concrete is not acceptable.

3.1.3.7 Blocking showing evidence of excessive crushing, warping, cracking, checking, rotting, or damage from dogging, loss of contact at edges caused by cracking or unequal shrinkage or deterioration to an extent no longer being capable of supporting a prescribed load over full bearing areas is not acceptable and shall be replaced.

3.1.4 Crib and Brace Blocking: Ensure that crib and brace blocking are in accordance the requirements noted below using the drawing applicable to the docking facility.

3.1.4.1 Blocks in excess of 6 feet in height but below 8 feet 6 inches in height shall be considered high blocks. Blocks which are 8 feet 6 inches or greater in height shall be considered extra-high blocks.

3.1.4.2 High keel blocks shall be cribbed or butted in the aft one-third and forward one-third of the keel block line. All extra-high keel blocks shall be cribbed or butted.

3.1.4.3 Side blocks above 6 feet in height (measured from the bottom of the block to the highest corner of the soft cap) shall be tied together longitudinally in pairs by means of steel bracing, or joined by cribbing, or be constructed as individual pyramids. If the side blocks are haul blocks, they must be hauled in pairs.

3.1.5 Setting and Alignment of Docking Blocks: Furnish, set, record, and align the blocks in accordance with the Government-Approved Contractor blocking plan. Prepare dry dock for inspection a minimum of 5 workdays prior to the proposed docking. Notify and accomplish an inspection of the blocking with the GDO in a timeframe sufficient to make blocking changes. Develop documentation of the inspection showing blocking, sighting marks on dock coping relative to block locations, spacing, offsets, heights, and shaping to the GDO at least 4 hours prior to flooding the dry dock. Immediately prior to flooding the dry dock, perform a final inspection with the GDO to ensure that no loose trash, wood, equipment, and other extraneous matter could float up during flooding and interfere with/delay the docking operation.

3.1.6 Docking Conference: Perform a docking conference attended by the GDO and SUPERVISOR, the Contractor's Dock Master, the pilot who will be present during the docking evolution and required dive personnel within 4

workdays prior to the scheduled docking of the vessel. The Dock Master shall discuss the time of docking, tide situation, weather conditions, tug and pilot arrangements, line handling, electrical shore power, grounding, safety issues, safety concerns, other services, and procedures.

3.1.7 Docking the Vessel: Dry dock the vessel without strain. Immediately after docking, electrically connect the vessel hull at the bow and stern of vessel for protection against lightning and other static charges in accordance with the requirements of NSTM Chapter 074 Volume One.

3.1.7.1 Notify the GDO a minimum of 2 workdays prior to actual docking and undocking of vessel.

3.1.7.1.1 Contact the SUPERVISOR for current GDO contact information.

3.1.7.2 Dry dock facilities shall have an alternate / back up electrical power generating and distributing source to furnish all electrical requirements for docking and undocking of vessel, in the event of a loss of main power source.

3.1.8 Post Docking: Prepare and inspect vessel underwater hull surfaces.

3.1.8.1 Immediately after docking, perform a visual inspection of the underwater portions of the vessel with the GDO for damage and deterioration. Inspect the fit on the blocks and furnish necessary shimming between the blocking and the hull to prevent hull movement and to ensure 100 percent block contact with the hull structure.

3.1.8.2 Within 8 hours of docking, wash the underwater hull and fittings with fresh water at a minimum of 3,000 PSIG nozzle pressure to remove dirt, slime, marine growth, fouling and other foreign substances.

3.1.8.3 Comply with federal, state, and local laws and ordinances for disposal of wash water and debris removed from underwater portions of the vessel.

3.1.9 Undocking the Vessel: After completion of work, undock the vessel at a time and date agreed upon by the GDO and the Contractor. Perform an undocking conference similar to the Docking Conference of 3.1.6 within 3 workdays of undocking. Immediately prior to flooding the dry dock, perform an inspection with the GDO to ensure that no loose trash, wood, equipment, or other

extraneous matter could float up during flooding and interfere with/delay the undocking operation.

3.1.9.1 Immediately after the hull penetrations are submerged and before the vessel lifts off the blocks-- stop flooding the dry dock. Accomplish a watertight integrity check of the hull with the SUPERVISOR. Continue flooding of the dock when directed by the GDO.

3.1.9.2 After the vessel has cleared the blocks, perform a visual inspection of the underwater hull surfaces to ensure that no soft caps are adhered to the hull surfaces. Perform a visual inspection of the docking blocks to account for all soft caps and to ensure that no blocks are adhered to the underwater hull structure of the vessel.

3.1.10 Docking Observer: The Docking Observer is the Government official certified by the Government as qualified to execute the duties of a Docking Observer. The presence of the Docking Observer is for protecting the Navy's interests, not to give direction to the Contractor. However, the Docking Observer shall have the authority to stop the docking, undocking, or launching evolution when the vessel safety or other Navy interests are jeopardized. This authority does not imply limitations on or responsibility for the actions of the Contractor's Dock Master.

3.1.11 Final Docking Report: Develop a final docking report (NAVSEA 9997/1 through 9997/5, as necessary) upon completion of undocking. Electronic versions of the final docking report, in portable document format (PDF) and MSWord format are available from the GDO upon request. **(REQUIRED REPORT CDRL A026)**

3.2 Hull Repairs:

3.2.1 Identify Distinguishing Figures and Markings: Identify and record existing distinguishing underwater body, hull, and freeboard color schemes, markings, and figures in sketch form listing colors, locations, sizes, and distinguishing marking(s), and figure(s) information, using drawing S2804-921819 Rev C for guidance.

3.2.2 Zinc Anodes: Remove existing and install new zinc anodes and associated CRES 316 fasteners, using drawings 100-7281566 Rev D and 803-921865 Rev T for guidance.

3.2.2.1 Install new QTY (107 EA) hull zinc anodes conforming to the requirements of Type ZHC-23 on the underwater hull and body of the vessel.

3.2.2.2 Identify and record exact location of each zinc anode.

3.2.3.2.1 List and identify corresponding specific measurements, locations, and/or areas in relationship to fixed hull structural members and docking block locations, or with measurements from suitable hull reference points.

3.2.3.2.2 A sketch of the underwater body and freeboard plating surfaces with appropriate data and measurements is an acceptable alternative. (REQUIRED REPORT CDRL A027)

3.2.3.3 Accomplish an electrical resistance test for each new zinc anode in accordance with General Note 4 of drawing 803-921865 Rev T. (REQUIRED REPORT CDRL A028)

3.2.3.4 New zincs shall remain free of preservation coatings.

3.3 **Structural Repairs:**

3.3.1 Ultrasonic Inspection: Accomplish a UT measurement inspection of the Hull, Main Deck, and Structural Member surfaces using drawings 100-5283618-A, 100-5283619-A, 100-7281566 Rev D, and 145-6773239 for guidance.

3.3.1.1 UT inspection locations will be identified by the ABS surveyor.

TABLE ONE – Anticipated UT Locations	
Location	Estimated Number of UT Readings
Main Deck	90 each
Bottom Plate	75 each
Stern Log	1 each
Head Log	1 each
Stbd Hull Plate (incl. bilge and shear strakes)	60 each
Port Hull Plate (incl. bilge and shear strakes)	60 each

Frame “X”, Tank “X”, Girth Belt (incl. structural members)	45 each
Frame “Y”, Tank “Y”, Girth Belt (incl. structural members)	45 each
Frame “Z”, Tank “Z”, Girth Belt (incl. structural members)	45 each
Other (TBD)	28 each

3.3.1.2 Measure and record in columnar form, 450 sites using TABLE TWO below for guidance. **(REQUIRED REPORT CDRL A029)**

TABLE TWO			
Original Plating Thickness (inch)			Minimum Plate Thickness
Fractional	Decimal	Pound/SF	25% Wastage
1/16	0.063	2.55	0.047
1/8	0.125	5.10	0.094
3/16	0.188	7.65	0.141
1/4	0.250	10.20	0.188
5/16	0.313	12.75	0.235
3/8	0.375	15.30	0.281
7/16	0.438	17.85	0.329
1/2	0.500	20.40	0.375
9/16	0.563	22.95	0.422
5/8	0.625	25.53	0.469
3/4	0.750	30.63	0.563
7/8	0.875	35.74	0.656
1	1.000	40.84	0.750

3.3.1.2.1 List in one column the design thickness.

3.3.1.2.2 Record in a separate column if reading is taken over an existing doubler or insert plate.

3.3.1.2.3 List in another column, next to the design thickness, the actual readings taken.

3.3.1.2.4 List and identify corresponding specific measurements, locations, and/or areas in relationship to structural members or with measurements from suitable hull reference points.

3.3.1.2.5 A sketch of the Main Deck plating with appropriate data is an acceptable alternative.

3.3.1.2.6 Calculate, identify, and highlight measurements indicating a difference (wastage) greater than or equal to 25 percent of the designed plate thickness, using TABLE TWO for guidance.

3.4 **Tanks/Voids:**

3.4.1 Open, Pump, Clean, Ventilate, and Gas Free Tanks/Voids: Open, pump, clean, ventilate, gas free, maintain dry, and Safe for Workers/Safe for Hot Work, as applicable, each location listed in TABLE THREE; retain access cover fasteners for re-install, using drawing 145-6773239 for guidance.

3.4.1.1 Meter, measure, and record actual amounts of residual liquids (in gallons) and solids (in pounds) removed from each location listed in TABLE THREE. It is anticipated that the vessel will be delivered to the Contractor with tanks/voids at low suction and that an estimated cumulative total of 1,000 Gallons of residual liquids and an estimated cumulative total of 50 pounds of residual solids will be removed from the vessel after delivery to the Contractor.

TABLE THREE – Tank Designations			
TK-010	TK-030	TK-070	TK-092
TK-011	TK-040	TK-080	TK-100
TK-012	TK-050	TK-090	TK-101
TK-020	TK-060	TK-091	TK-102

3.4.2 Preliminary Air Test: Accomplish preliminary air test(s) at each location listed in TABLE THREE at 2 pounds per square inch (PSI). Test pressure shall remain stable and constant during the test. Apply a soap solution to the opposite side of boundaries under test and inspect to determine leakage sites.

3.4.2.1 Install temporary expandable FME blanks and/or plugs, painted blaze orange, in each hull opening and each designed tank/void surface penetration(s) required to isolate each tank/void under test from adjacent tanks/voids. Record and identify the location of each FME blank and/or plug installed, including each corresponding and specific measurement(s), location(s), and/or area(s) in relationship to structural members or from suitable hull reference points. Remove each temporary FME blank and/or plug, record removal date, and restore each tank/void to its original configuration upon final completion of testing, blasting, and preservation operations.

3.4.3 Tank/Void Visual Inspection (VT): Accomplish a VT inspection of 100 percent of interior surfaces of each location listed in TABLE THREE for structural deformation, including each cleaned gasket mating surface and threaded surfaces, for deterioration, and/or defects in coatings, using drawings 100-5283618-A, 100-5283619-A, 100-7281566 Rev D, 114-5322269-B, and 145-6773239 for guidance. **(REQUIRED REPORT CDRL A030)**

3.4.3.1 Search for, mark, identify, and record defective (actual and suspect) areas showing signs of cracks, pitting, corrosion, deterioration, damage, distortion, and unusual wear including all sounding tubes, tank vents, overflows, piping, structural members, and access covers.

3.4.4 Install Access Covers: Accomplish access cover installation for each location listed in TABLE THREE, installing new MIL-G-1149, Class 2 Rubber-Synthetic HH-P-151, Rubber-Cloth Insert, 1/8-inch thickness gasket using retained fasteners.

3.4.4.1 Template exact sizes, locations, and configurations from existing vessel conditions.

3.4.4.2 Install anti-seize sealant compound conforming to the requirements of CID-A-A-59491 prior to installation of covers.

(V)(G) “FINAL AIR TEST”

3.4.5 Final Air Test: Accomplish final air test for each location listed in TABLE THREE after all repair work is completed. Pressure shall stabilize 15 minutes prior to conducting test. Hold pressure at 2 PSI for 10 minutes with no drop in pressure. Apply a soap solution to the opposite side of boundaries under test and inspect to determine leakage sites. Allowable leakage: None.

(REQUIRED REPORT CDRL A031)

3.4.5.1 Install temporary expandable foreign material exclusion (FME) blanks and/or plugs, painted blaze orange, in each hull opening and each designed tank/void surface penetration(s) required to isolate each tank/void under test from adjacent tanks/voids. Record and identify the location of each FME blank and/or plug installed, including each corresponding and specific measurement(s), location(s), and/or area(s) in relationship to structural members or from suitable hull reference points. Remove each temporary FME blank and/or plug, record removal date, and restore each tank/void to its original configuration upon final completion of testing, blasting, and preservation operations.

3.4.5.2 Upon successful completion of final air tests, remove each temporary FME blank and/or plug installed in 3.4.5.1.

(V)(G) “TANK/VOID CLOSEOUT INSPECTION”

3.4.6 Tank/Void Closeout Inspection: Schedule and accomplish a tank/void closeout inspection of each location listed in TABLE THREE with the SUPERVISOR prior to final close of each tank/void.

3.4.6.1 Notify the SUPERVISOR via email, 2 workdays prior to accomplishing each tank/void closeout inspection(s).

3.4.7 American Bureau of Shipping (ABS) and Coast Guard (CG) Survey Support: Furnish qualified and experienced personnel, facilities, and the necessary equipment to facilitate the ABS and CG inspectors for the access and inspection of all tanks/voids, main deck, and exterior hull, including:

3.4.7.1 Supporting employees to accomplish the inspection of all tanks/voids.

3.4.7.2 Tanks shall be gas free tested and maintained, certified as safe for entry eight (8) hours prior to and throughout all ABS inspections.

3.5 **Main Deck Repairs:** Not Used

3.6 **Superstructure:** Not Used

3.7 **Main Propulsion:** Not Used

3.8 **Steering Repairs:** Not Used

3.9 **Machinery Systems:** Not Used

3.10 **Electrical Systems:** Not Used

3.11 **Piping Repairs:** Not Used

3.12 **Weight Handling Systems:** Not Used

3.13 **Dock And Sea Trials:** Not Used

4.0 **GOVERNMENT FURNISHED MATERIAL (GFM):**

	Quantity	UI	Name of Part/Description	Pc No.	Ref No.	National Stock No.	Paragraph No.
4.1	N/A	N/A	N/A	N/A	N/A	N/A	N/A

5.0 **DATA SUBMISSION LIST:** The below listed reports shall be submitted to the SUPERVISOR.

DATA SUBMISSION LIST			
CA = Contract Award, DD = Drydocking, PCD – Project Completion Date, PM – Production Meeting, PSD = Production Start Date			
CDRL No.	Paragraph No.	REPORT TYPE	Date Required
A001	1.5.1	Production Performance Schedule	Initial, PSD (- 5) Days; Weekly, PM (- 24) Hours
A002	1.5.2	SubKTR List	CA + 5 days, Revision + 3 of change.
A003	1.5.4	Welder Qualification List	CA + 5 days, Revision + 3 of change.
A004	1.5.5	Manufacturer's Product Data Sheet	CA + 5 Days
A005	1.5.6	Safety Data Sheets	CA + 5 Days
A006	1.5.7	Key KTR Personnel	CA + 5 days, Revision + 3 of change.
A007	1.5.8	KTR Insp and QC System	CA + 5 days, Revision + 3 of change.
A008	1.5.9	KTR Environmental Protection, Hazardous Waste Management Plan	CA + 5 days, Revision + 3 of change.
A009	1.7.8	Arrival Conditions Photos	Inspection +2 Days
A010	1.11	CFR	Initial, PSD (- 5) Days. Supporting Each Report, Inspection, Test
A011	1.13.2	Incident Report	Investigation Complete +24 Hrs.
A012	1.16.1	Paint Preservation Appendices	SSPC SP1 + 1 Day; Surface Profiles + 1 Day; Surface Conductivity + 1 Day; Topcoat Cure + 1 Day
A013	1.16.2.1	Data Logger	Inspection +2 Days

DATA SUBMISSION LIST			
CA = Contract Award, DD = Drydocking, PCD – Project Completion Date, PM – Production Meeting, PSD = Production Start Date			
CDRL No.	Paragraph No.	REPORT TYPE	Date Required
A014	1.17	Hazardous Waste Reports	PCD + 30 Days
A015	1.18.2	Contractor Furnished Material (CFM) Report	CA +30
A016	1.18.3	Material Certifications - ABS	Mtrl Receipt + 2 Days
A017	1.22.1	Gas Free	PSD + 8 Days; Revisions + 2 Days
A018	1.22.1.1	Third Party Rescue Plan	CA +5 Days
A019	1.22.2	Marine Chemist Gas Free Certificate & Competent Person List	Inspection +2 Days
A020	1.23.1	Test and Inspection Plan	Weekly, PM (- 24) Hours; Final +24 Hrs. after All Testing Complete
A021	1.23.5.2	(G) Points Insp Reports	Inspection +2 Days
A022	1.27	Heavy Weather Plan	CA +5 Days
A023	1.28.3	FME Log	Inspection +2 Days
A024	3.1.2	Docking Procedure	DD (-5) Days
A025	3.1.3	Blocking Plan	DD (-5) Dats
A026	3.1.11	Final Docking Report	Undocking +5 Days
A027	3.2.3.2.2	Zinc Anode Location Map	Undocking (-5) Days
A028	3.2.3.3	Zinc Anode Electrical Resistance Test	Undocking (-5) Days
A029	3.3.1.2	UT Inspection Report(s)	Inspection +2 Days
A030	3.4.3	Tanks/Voids VT	Inspection +2 Days
A031	3.4.5	Final Air Test	Inspection +2 Days
A032	1.14.3.2	ABS Inspection Reports	Inspection + 2 Days